ISEN/DAEN 427 Lecture 5

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Bayesian Inference with Beta Prior and Binomial Likelihood
Cue Combination

BM Ch. 5, 6

Bayesian Inference with Binomial Likelihood

- Assume observations $x_1, \ldots, x_N$ are binary $x_i \in \{0, 1\}$ and $s \in [0, 1]$.
- Given a likelihood $\mathbb{P}(x_i|s)$, conditional independence implies:

$$\begin{aligned}
\mathbb{P}(x_1, \ldots, x_N|s) &= \prod_{i=1}^{N} \mathbb{P}(x_i|s) \\
&= \prod_{i=1}^{N} s^{x_i}(1-s)^{1-x_i} \\
&= s^{\sum_{i=i}^{N} x_i}(1-s)^{N-\sum_{i=i}^{N} x_i}
\end{aligned}$$

where $\sum_{i=i}^{N} x_i$ is the total number of ones ("successes") and $N - \sum_{i=i}^{N} x_i$ are the total number of zeros ("failures").

Bayesian Inference with Binomial Likelihood

- The likelihood can alternatively be written in terms of the Binomial distribution for $y = \sum_{i=i}^{N} x_i$:

$$\mathbb{P}(y|s) = \binom{N}{y} s^{y} (1-s)^{N-y}$$

  where $y \in \{0, 1, \ldots, N\}$.
- Recall that

$$\mathbb{E}[y] = Ns \qquad \mathrm{Var}[y] = Ns(1-s)$$

Prior: Beta distribution

- The beta distribution with parameters $\alpha > 0$ and $\beta > 0$ is of the form:

$$\mathbb{P}(s) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} s^{\alpha-1}(1-s)^{\beta-1}$$

  where $\Gamma(\cdot)$ function is defined as:

$$\Gamma(x) \triangleq \int_0^\infty u^{x-1} e^{-u} du$$

  and for positive integer n, $\Gamma(n) = (n-1)!$

- The mean and variance: $\mathbb{E}[s] = \frac{\alpha}{\alpha+\beta}$ and $\mathrm{Var}[s] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

Learning as Inference

Assume the a priori distribution is beta with parameters $\alpha > 0$ and $\beta > 0$ and $y = \sum_{i=i}^{N} x_i$ then

$$\begin{aligned}
\mathbb{P}(s|y) &= \kappa \mathbb{P}(y|s)\mathbb{P}(s) \\
&= \kappa \binom{N}{y} s^{y}(1-s)^{N-y} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} s^{\alpha-1}(1-s)^{\beta-1} \\
&= \kappa \binom{N}{y} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} s^{y+\alpha-1}(1-s)^{N+\beta-y-1}
\end{aligned}$$

where

$$\kappa = \frac{1}{\int_0^1 \mathbb{P}(y|s)\mathbb{P}(s)ds} = \frac{1}{\mathbb{P}(y)}$$

Computing the Posterior

$$\begin{aligned}
\mathbb{P}(y) &= \int_0^1 \mathbb{P}(y|s)\mathbb{P}(s)ds \\
&= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \binom{N}{y} s^y (1-s)^{N-y} s^{\alpha-1}(1-s)^{\beta-1} ds \\
&= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \binom{N}{y} s^{y+\alpha-1}(1-s)^{N-y+\beta-1} ds \\
&= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha')\Gamma(\beta')}{\Gamma(\alpha'+\beta')} \binom{N}{y}
\end{aligned}$$

where $\alpha' = y + \alpha$ and $\beta' = N - y + \beta$

Computing the Posterior

- After recording y ones ("successes") out of $N > 1$ the posterior distribution is beta with parameters α' and β', i.e.:

$$\mathbb{P}(s|y) = \frac{\Gamma(\alpha' + \beta')}{\Gamma(\alpha')\Gamma(\beta')} s^{\alpha' - 1}(1 - s)^{\beta' - 1}$$

- Thus,

$$\mu_{post} = \frac{\alpha'}{\alpha' + \beta'} = \frac{y + \alpha}{N + \alpha + \beta}$$

$$\begin{aligned} \sigma_{post} &= \frac{\alpha'\beta'}{(\alpha' + \beta')^2(\alpha' + \beta' + 1)} \\ &= \frac{(y + \alpha)(N - y + \beta)}{(N + \alpha + \beta)^2(N + \alpha + \beta + 1)} \end{aligned}$$

Computing the Posterior

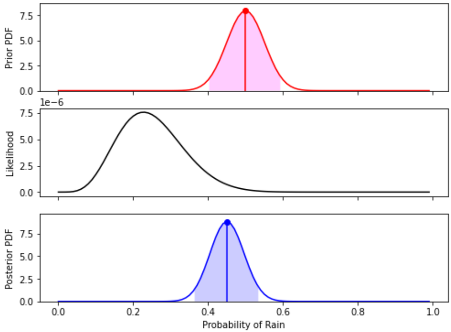

Figure 1: Posterior with Beta Prior and Binomial Likelihood

$s = \mathbb{P}(\text{“rain”})$

Connection to Reinforcement Learning

- With $\alpha = \beta = 1$, an additional $N+1$ observation leads to the posterior mean

$$\mu_{post,N+1} = \frac{y + x_{N+1} + 1}{N + 1 + 2}$$

- A recursive update equation for the posterior mean

$$\begin{aligned}
\mu_{post,N+1} &= \frac{y+1}{N+2}\frac{N+2}{N+3} + x_{N+1}\frac{1}{N+3} \\
&= \mu_{post,N}(1 - \frac{1}{N+3}) + x_{N+1}\frac{1}{N+3} \\
&= \mu_{post,N} + \frac{1}{N+3}(x_{N+1} - \mu_{post,N})
\end{aligned}$$

Connection to Reinforcement Learning

- This equation resembles the *Rescorla-Wagner* rule in reinforcement learning.
- New information or "surprise" is $x_{N+1} - \mu_{post,N}$.
- The factor $\frac{1}{N+3}$ acts as the learning rate.

Cue Combination

- Inference based on two *different* types of measurements (cues).
- How should cues be processed in order to optimize inference ?
- Experimental evidence indicates the human brain combines cues.

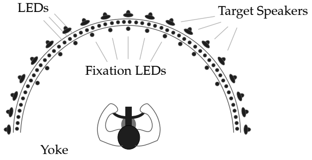

Figure 2: Testing Visual and Auditory Cue Combination

Cue Combination

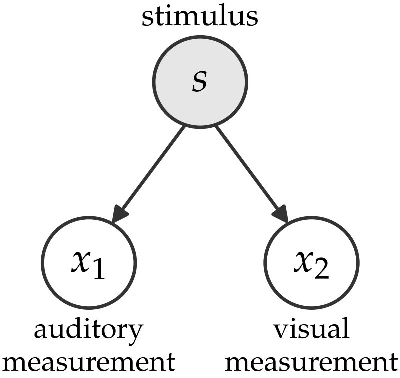

Figure 3: Generative Model

- Let x_1 and x_2 denote the observation values (cues). We assume

$$\mathbb{P}(x_1, x_2 | s) = \mathbb{P}(x_1 | s)\mathbb{P}(x_2 | s)$$

- Let us derive the posterior when

$$\mathbb{P}(x_i | s) = \frac{1}{\sqrt{2\pi\sigma_i}} \exp\left(-\frac{1}{2\sigma_i^2}(x_i - s)^2\right)$$

for $i = 1, 2$.

Posterior Distribution

- Similarly to derivation in Lecture 4, it can be shown that:

$$\mathbb{P}(s|x_1, x_2) = \frac{1}{\sqrt{2\pi\sigma_{post}}} \exp(-\frac{1}{2\sigma_i^2}(s - \mu_{post})^2)$$

with

$$\mu_{post} = \frac{\frac{1}{\sigma_s^2}}{\frac{1}{\sigma_s^2}+\frac{1}{\sigma_1^2}+\frac{1}{\sigma_2^2}}\mu_s + \frac{\frac{1}{\sigma_1^2}}{\frac{1}{\sigma_s^2}+\frac{1}{\sigma_1^2}+\frac{1}{\sigma_2^2}}x_1 + \frac{\frac{1}{\sigma_2^2}}{\frac{1}{\sigma_s^2}+\frac{1}{\sigma_1^2}+\frac{1}{\sigma_2^2}}x_2$$

$$\sigma_{post} = \frac{1}{\frac{1}{\sigma_s^2}+\frac{1}{\sigma_1^2}+\frac{1}{\sigma_2^2}}$$

Posterior Distribution

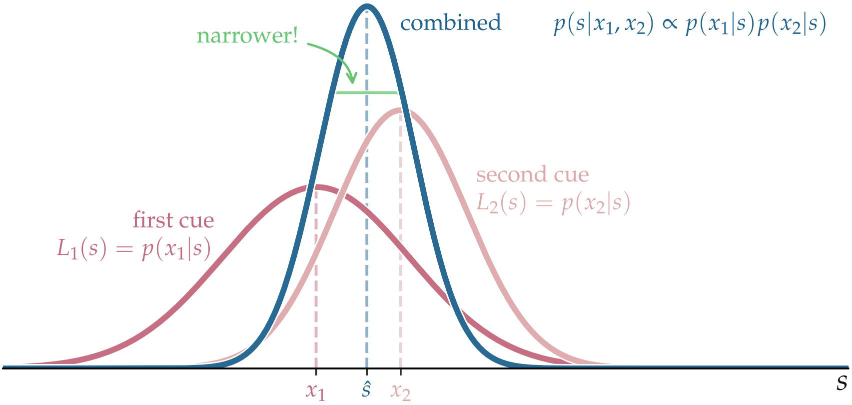

Figure 4: Posterior Distribution from Cue Combination